RECAP

MATILDE DOLFATO

Hyperparameters

Labels are generated as

$$y \leftarrow \operatorname{sign}\left(5 \sin\left(\frac{\pi\,(m+1)}{10}\,x\right) + \sigma\varepsilon\right), \qquad \varepsilon \sim \mathcal{N}(0,1),$$

with x sampled uniformly from N evenly spaced points of $(0, 10)$. The noise level σ is chosen so that the average flip probability $\frac{1}{N}\sum_j \Phi\left(-|h^*(\bar{x}_j)|/\sigma\right)$ equals the target percentage of flipped labels.

Parameter	Symbol	Value
Support size	N	10^9
Train size	n	500
Test size	n_{test}	$200,000$
Iterations	T	100
Crossings of h^*	m	10
Noise	σ	≈ 3.5, 20% flipped labels
Depths	d	$1, \ldots, \lfloor 4 \log_2 n \rfloor$
Constant, union bound	C_{union}	$6 \cdot 10^{-4}$
Constant, Rademacher–Massart	C_{radmass}	0.125
Constant, noise bound	C_{noise}	0.01
Seed		4

See Appendix A for the computation of the constants.

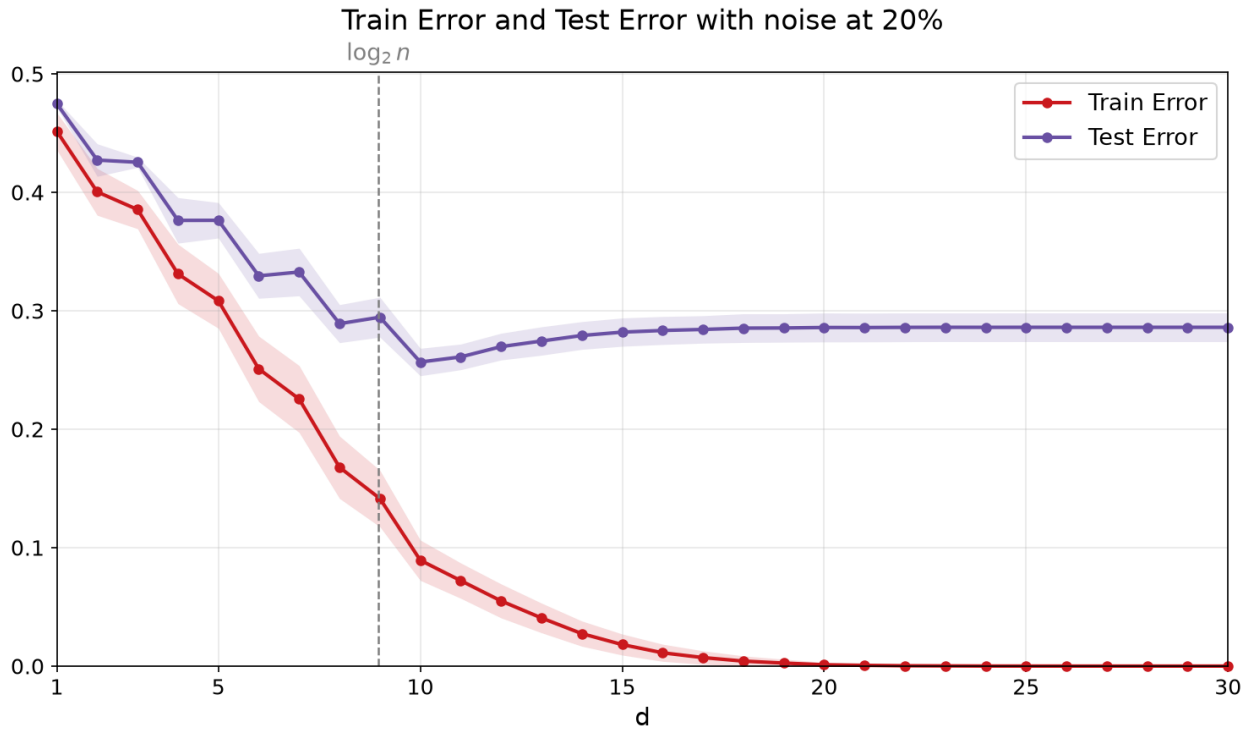


FIGURE 1. Train and test error as functions of the maximum depth d.

Hypotheses class

Consider the algorithm of binary decision trees with test at nodes $x \geq t$ for arbitrary $t \in \mathbb{R}$. When $\mathscr{H}_d$ is the class of functions represented by such trees of depth at most d,

$$|\mathscr{H}_d| = 2 \sum_{j=0}^{\min\{2^d-1, N-1\}} \binom{N-1}{j}$$

which yields

$$\log |\mathscr{H}_d| = (2^d - 1) \log \frac{eN}{2^d} = O(2^d) \qquad \text{for } d \ll \log_2 N$$

and

$$\log |\mathscr{H}_d| = N \log 2 \qquad \text{for } d \geq \log_2 N.$$

1. Union bound

Theorem 1.1. *Let ℓ be bounded, $\ell(\cdot) \leq M$ for some $M \in \mathbb{R}$. Then*

$$0 \leq \mathbb{E}[(R - \hat{R}_D)(\hat{h}_D)] \leq 2M\sqrt{\frac{\log|\mathscr{H}|}{2n}}$$

With the zero-one loss and $\mathscr{H} = \mathscr{H}_d$, we plot

$$6 \cdot 10^{-4} \cdot 2\sqrt{\frac{\log|\mathscr{H}_d|}{2n}}$$

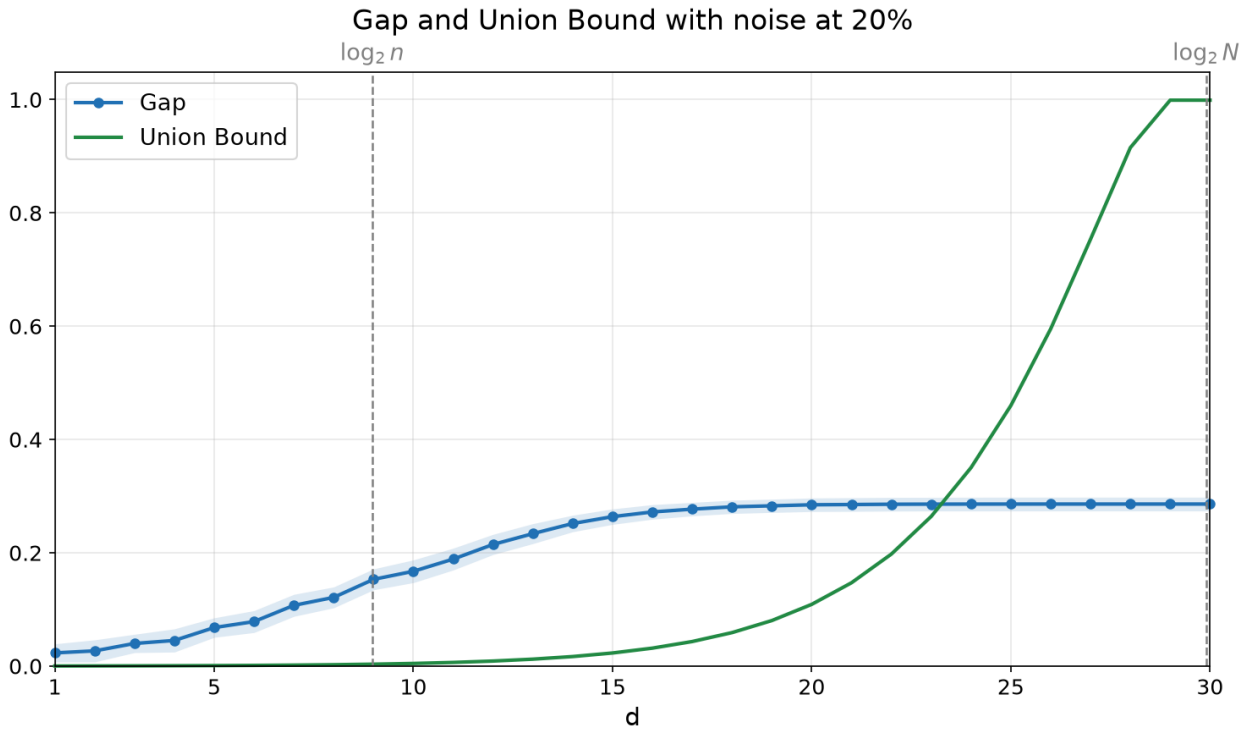


Figure 2. Union bound (Theorem 1.1) against the gap.

The generalization gap saturates around $d = 15$, slightly after the interpolation threshold $\lceil \log_2 n \rceil = 9$. The bound, instead, saturates at $\log_2 N = 30$. This bound does not capture well the shape of the generalization gap, in fact it is myopic to the algorithm. Indeed, if we didn't discretize the support, i.e. we let $N \to \infty$, the saturating value would be infinite.

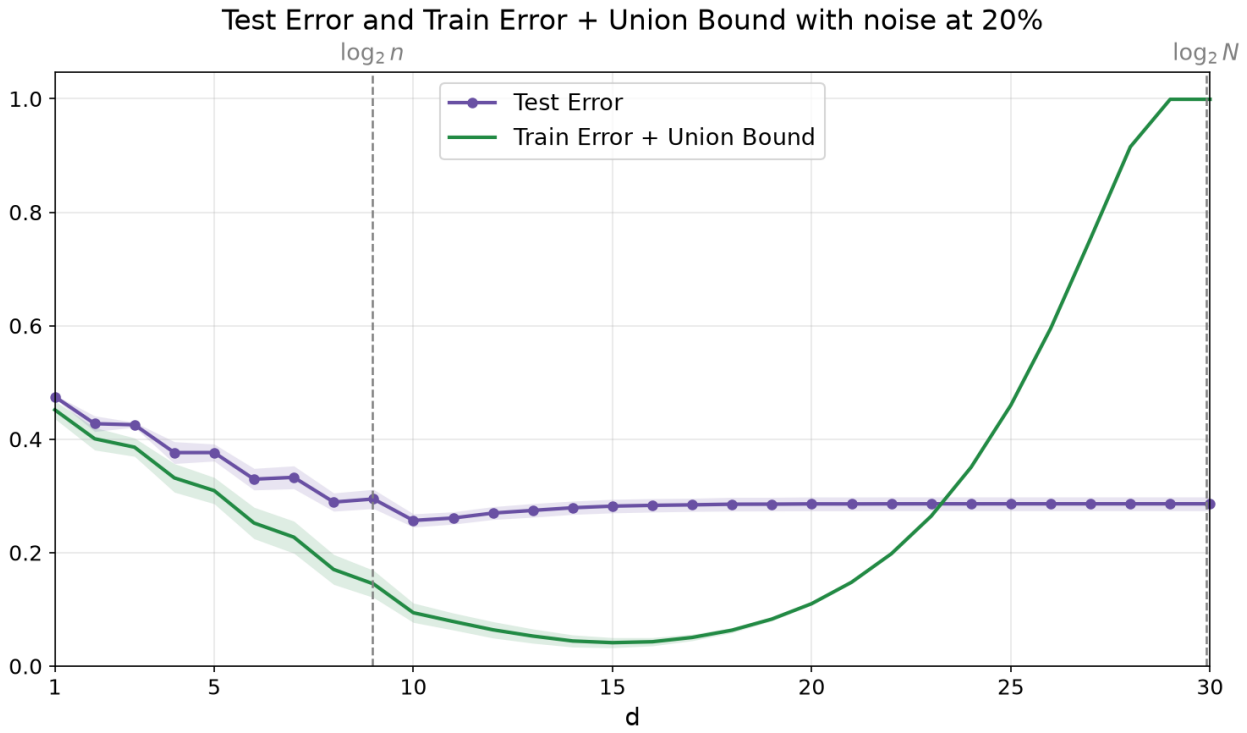


FIGURE 3. Test error against union bound (Theorem 1.1) plus train error.

2. Rademacher and Massart bound

Let

$$N_D(\mathscr{H}) \coloneqq |\{(\ell_h(X_1, Y_1), \ldots, \ell_h(X_n, Y_n)) : h \in \mathscr{H}, D = ((X_i, Y_i))_{i=1}^n\}|$$

be the cardinality of the set of loss vectors for the hypotheses in $\mathscr{H}$, which is a random variable as it depends on the dataset D.

Theorem 2.1 (Rademacher and Massart bound)**.** *Let ℓ be the zero-one loss. Then*

$$\mathbb{E}[(R - \hat{R}_D)(\hat{h}_D)] \leq 2\sqrt{\frac{2 \log \mathbb{E}[N_D(\mathscr{H})]}{n}}$$

Lemma 2.2 (Counting)**.** *Let ℓ be the zero-one loss. Then*

$$N_D(\mathscr{H}) \leq \min\{|\mathscr{H}|, 2^n\}$$

surely.

Substituting Lemma 2.2 into Theorem 2.1, we obtain the following.

Corollary 2.3. *Let ℓ be the zero-one loss. Then*

$$\mathbb{E}[(R - \hat{R}_D)(\hat{h}_D)] \leq \min\left\{2\sqrt{\frac{2 \log |\mathscr{H}|}{n}}, 2\sqrt{2 \log 2}\right\}$$

With the zero-one loss and $\mathscr{H} = \mathscr{H}_d$, we plot

$$0.125 \cdot \min\left\{2\sqrt{\frac{2 \log |\mathscr{H}_d|}{n}}, 2\sqrt{2 \log 2}\right\}$$

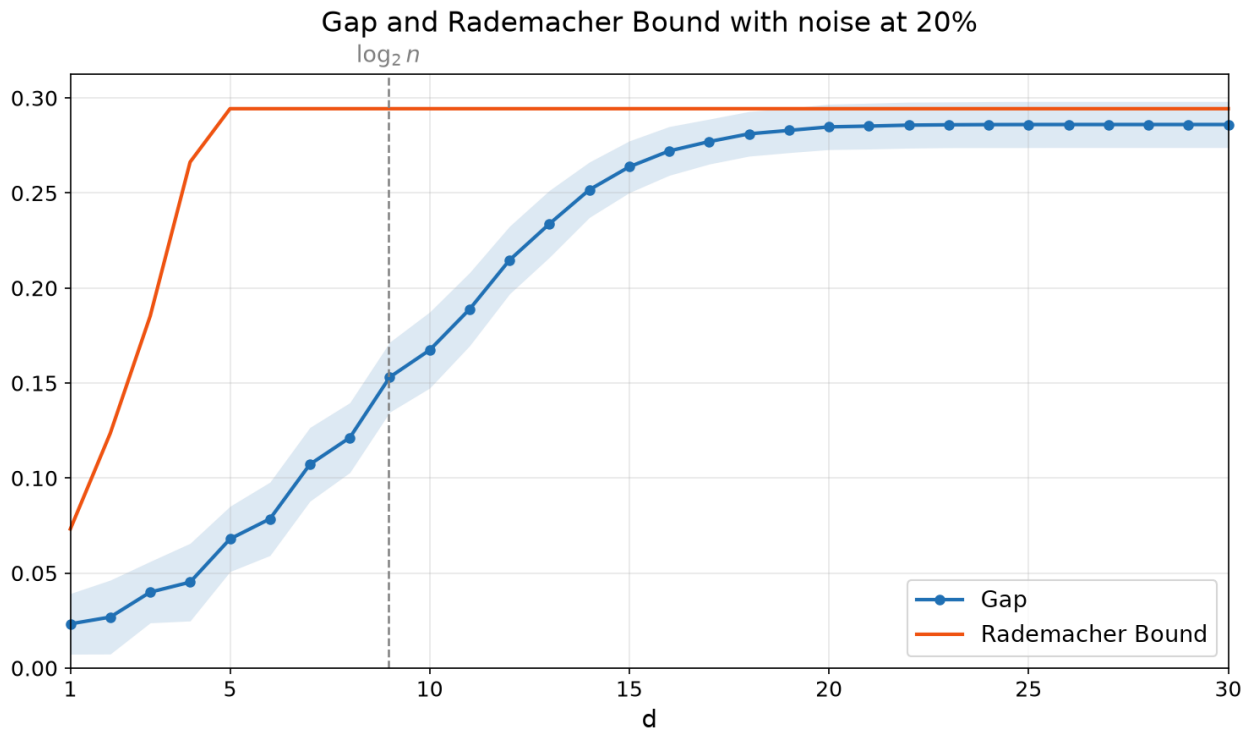


FIGURE 4. Rademacher and Massart bound (Corollary 2.3) against the gap.

The bound obtained in Corollary 2.3 better captures the curve of the test error, because it saturates from a lower depth. Specifically, for low values of the depth d, the minmum in Lemma 2.2 is attained by $|\mathscr{H}_d|$, while for a large enough depth the minimum becomes 2^n and the curve saturates. The depth at which the bound saturates is given by

$$\log |\mathscr{H}_d| = (2^d - 1) \log \frac{eN}{2^d} \geq n \quad \Rightarrow \quad d \geq 5$$

for a fixed $n = 500$.

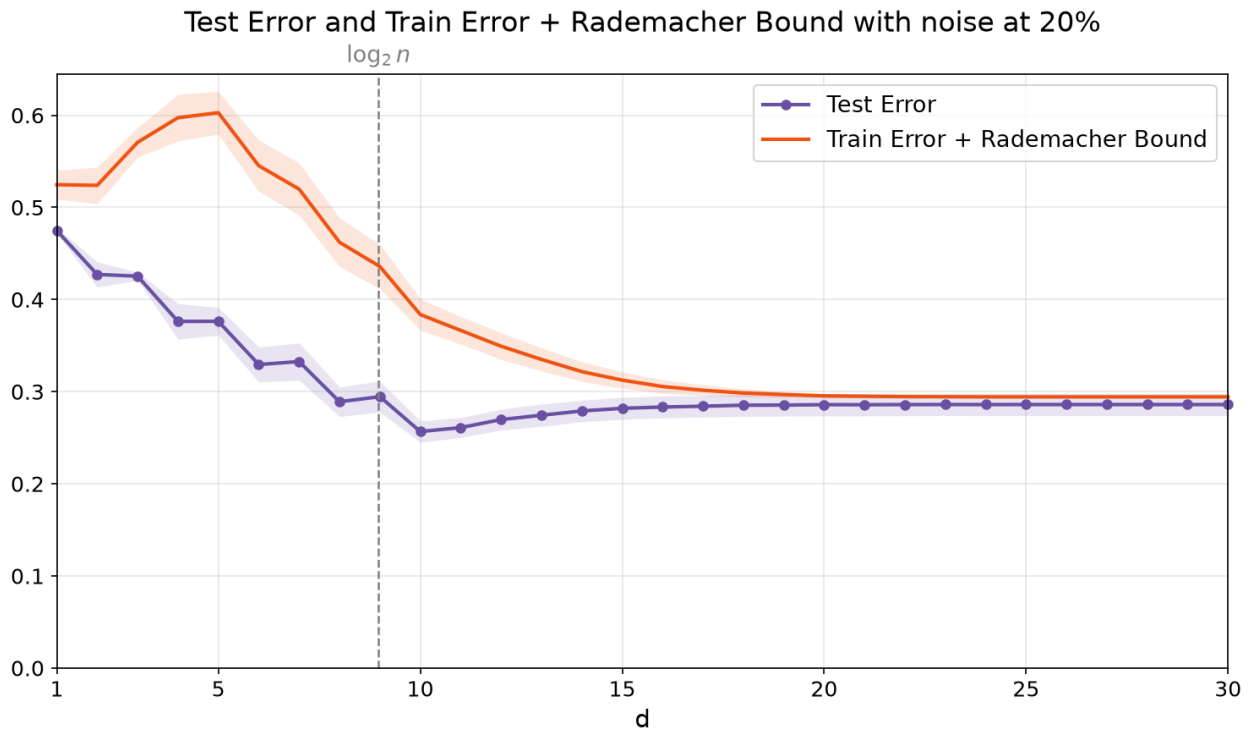


FIGURE 5. Test error against Rademacher and Massart bound (Corollary 2.3) plus train error.

Comparison with the union bound

Up to constants, the first term in the minimum of Corollary 2.3 is the union bound of Theorem 1.1. The two bounds coincide as long as $\log|\mathscr{H}_d| < n\log 2$, i.e. $|\mathscr{H}_d| < 2^n$, which with $n = 500$ and $N = 10^9$ holds for $d \leq 4$. From $d = 5$ onwards, the minimum is attained by the constant term and the Rademacher and Massart bound saturates, while the union bound keeps increasing up to $d = \log_2 N$.

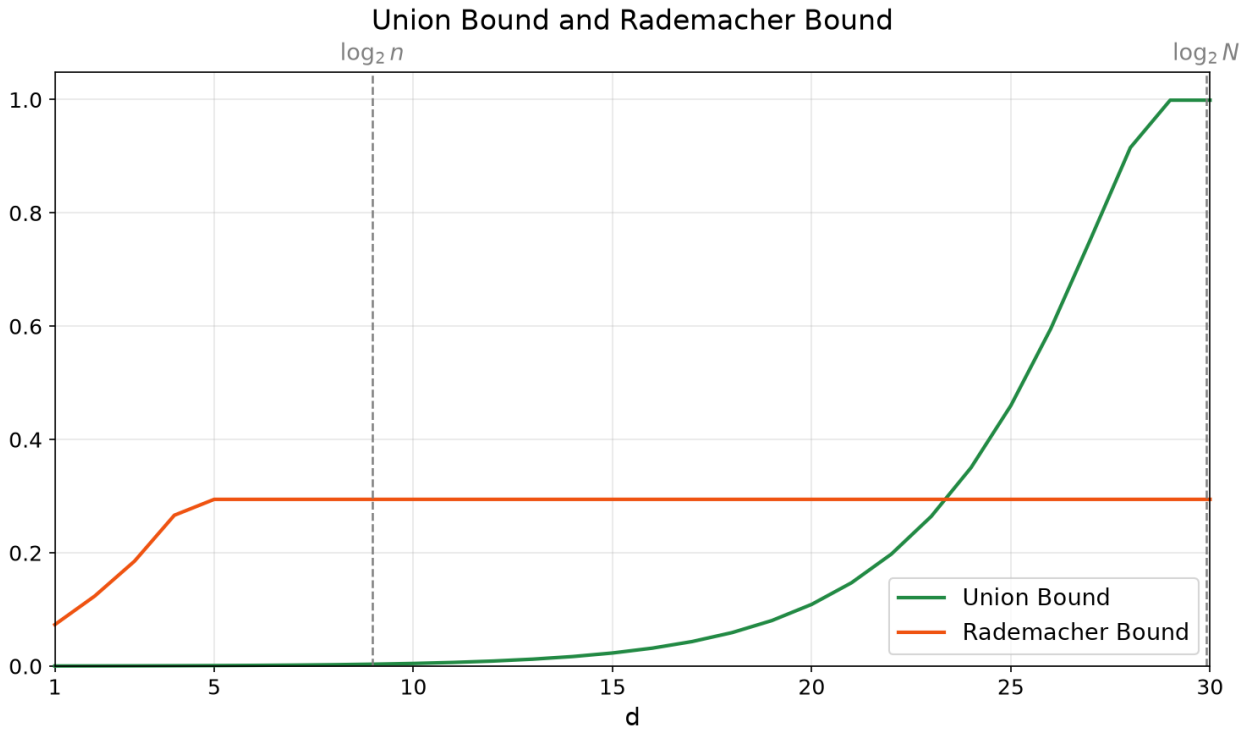


FIGURE 6. Union bound (Theorem 1.1) against Rademacher and Massart bound (Corollary 2.3).

Dependence on the noise

The bound of Corollary 2.3 does not depend on the noise, while the gap does: with 5%, 20% and 40% of flipped labels, it saturates at different levels, as shown in Figure 7.

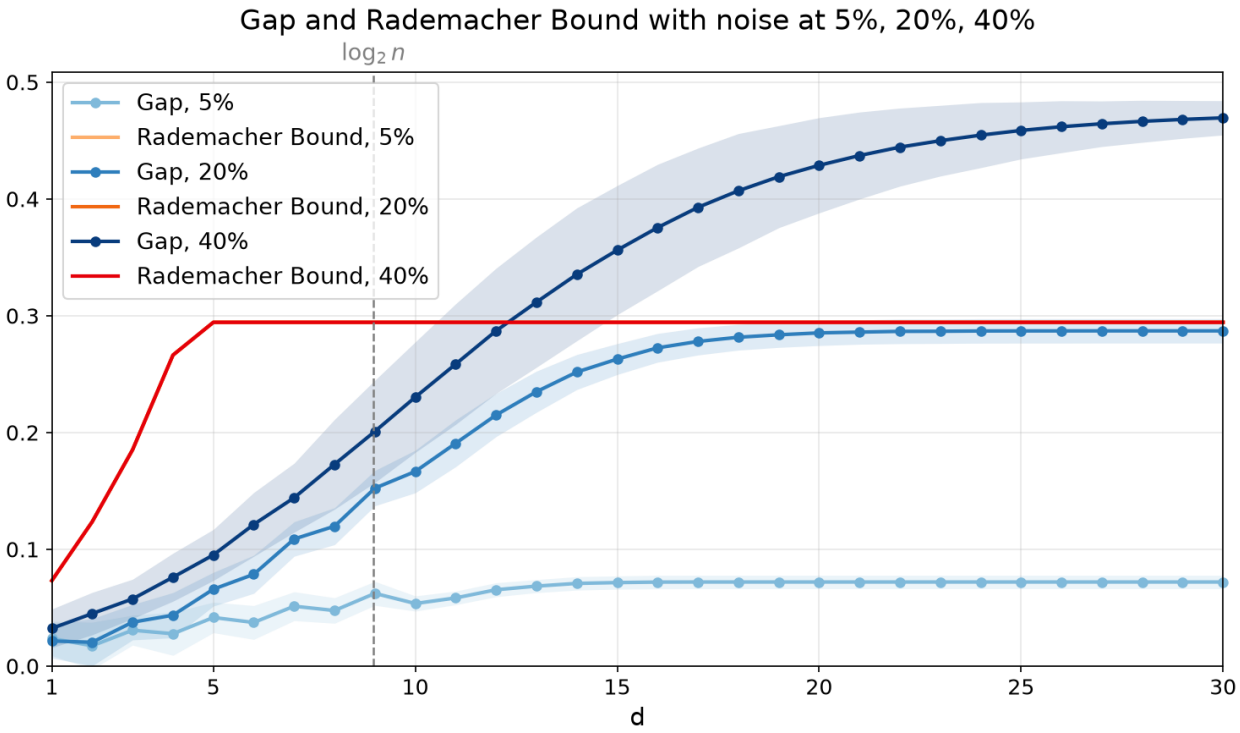


FIGURE 7. Rademacher and Massart bound (Corollary 2.3) against the gap for different noise levels.

3. Noise and approximation error bound

Let $R^*_{\mathscr{H}} = \min_{h\in\mathscr{H}} R(h)$ be the best risk, $R^* = \inf_{h:\mathcal{X}\to\mathcal{Y}} R(h)$ the Bayes risk and $A_{\mathscr{H}} = R^*_{\mathscr{H}} - R^*$ the approximation error.

Theorem 3.1 (Noise and approximation error bound)**.** *Let ℓ be the zero-one loss. Then*

$$\mathbb{E}\left[(R-\hat{R}_D)(\hat{h}_D)\right] \le \min\left\{4\sqrt{\frac{(R^*+A_{\mathscr{H}})\,(\log(4|\mathscr{H}|)+1)}{n}} + 20\,\frac{\log(4|\mathscr{H}|)+1}{n},\right.$$
$$\left. 4\sqrt{\frac{(R^*+A_{\mathscr{H}})\,((2n+2)\log 2+1)}{n}} + 20\,\frac{(2n+2)\log 2+1}{n}\right\}$$

With $\mathscr{H} = \mathscr{H}_d$, we plot

$$0.01\cdot\min\left\{4\sqrt{R^*_{\mathscr{H}_d}L_d}+20L_d,\ 4\sqrt{R^*_{\mathscr{H}_d}M}+20M\right\}, \qquad L_d = \frac{\log(4|\mathscr{H}_d|)+1}{n}, \quad M = \frac{(2n+2)\log 2+1}{n},$$

where $R^*_{\mathscr{H}_d} = R^* + A_{\mathscr{H}_d}$ is computed exactly for each d by dynamic programming.

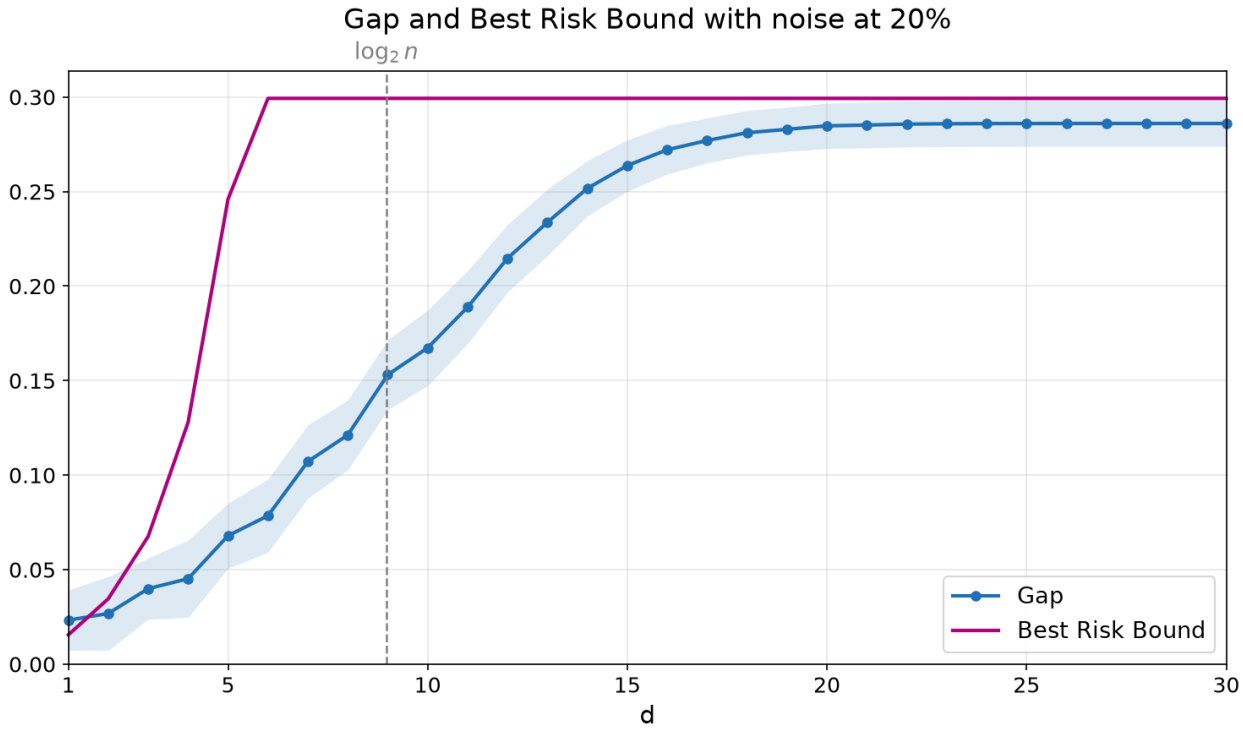


Figure 8. Noise and approximation error bound (Theorem 3.1) against the gap.

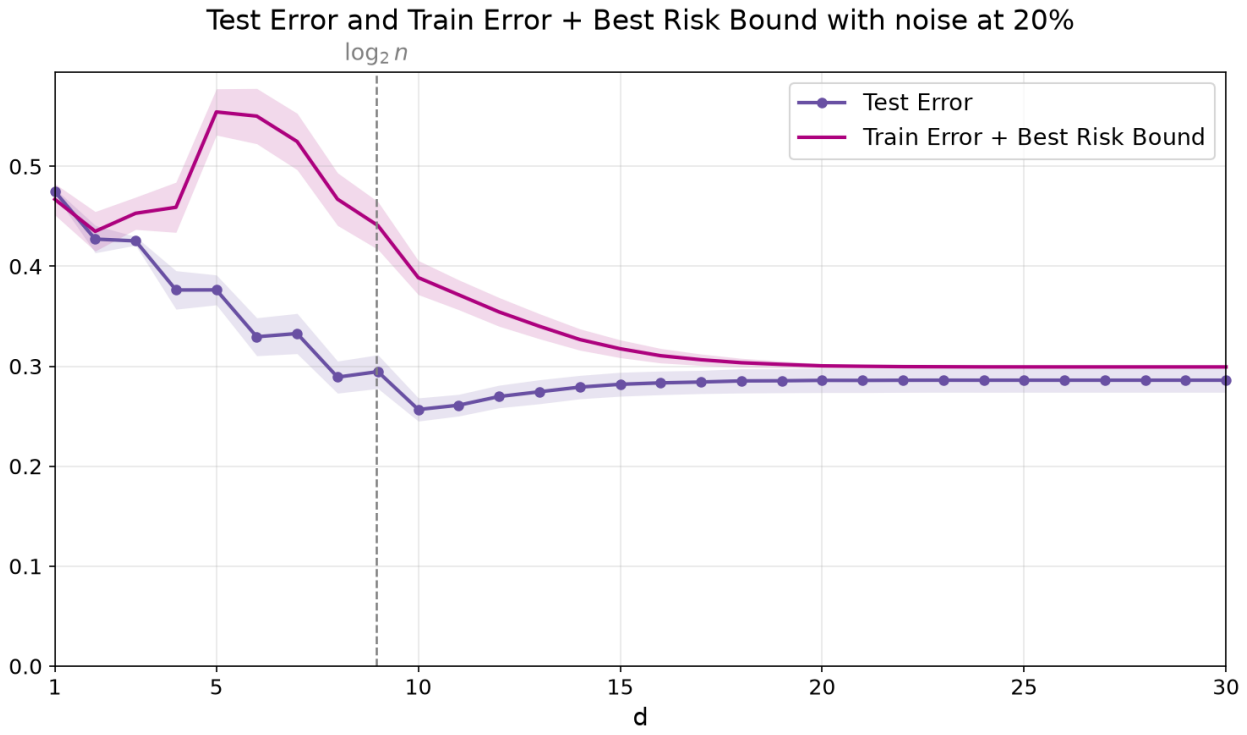


FIGURE 9. Test error against noise and approximation error bound (Theorem 3.1) plus train error.

Dependence on the noise

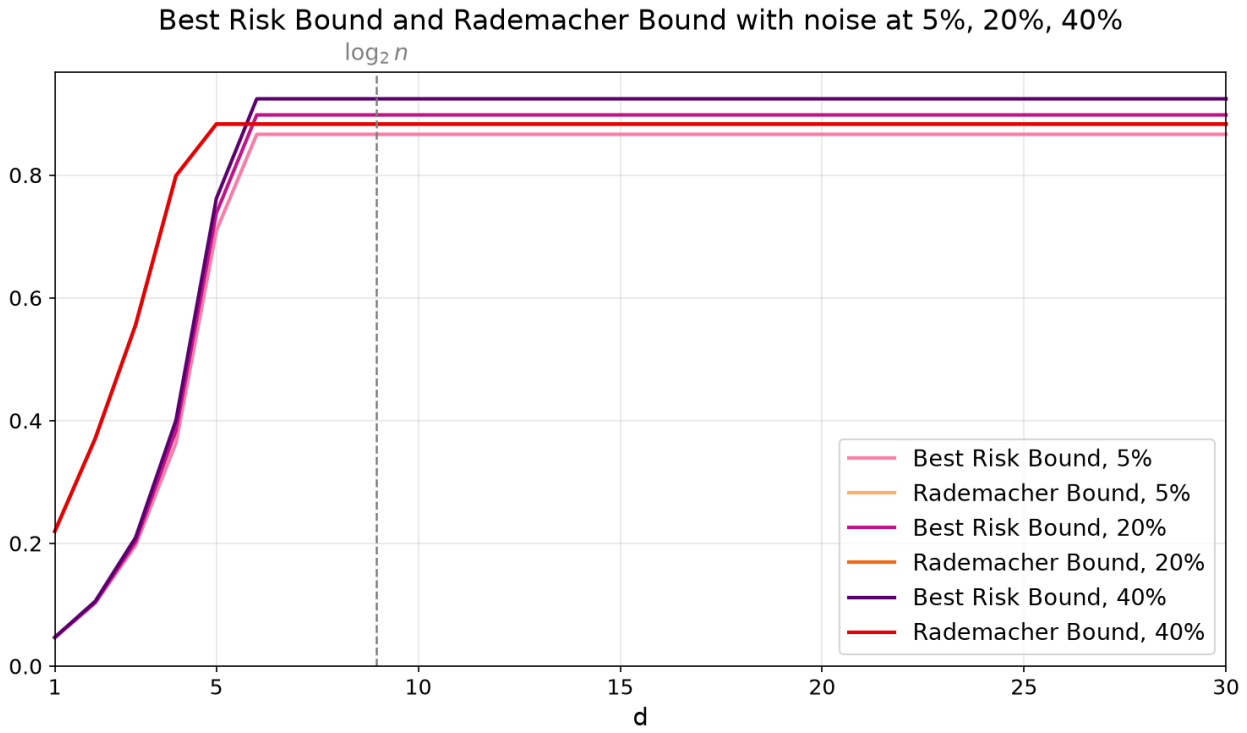


FIGURE 10. Noise and approximation error bound (Theorem 3.1) against Rademacher and Massart bound (Corollary 2.3) for different noise levels.

Figure 10 shows that the Rademacher and Massart bound does not change with the noise, while the noise and approximation error bound shifts upwards as the noise increases. Note that the constants used here are three times those used above (0.375 and 0.03), to better visualize the differences in the noise and approximation error bound.

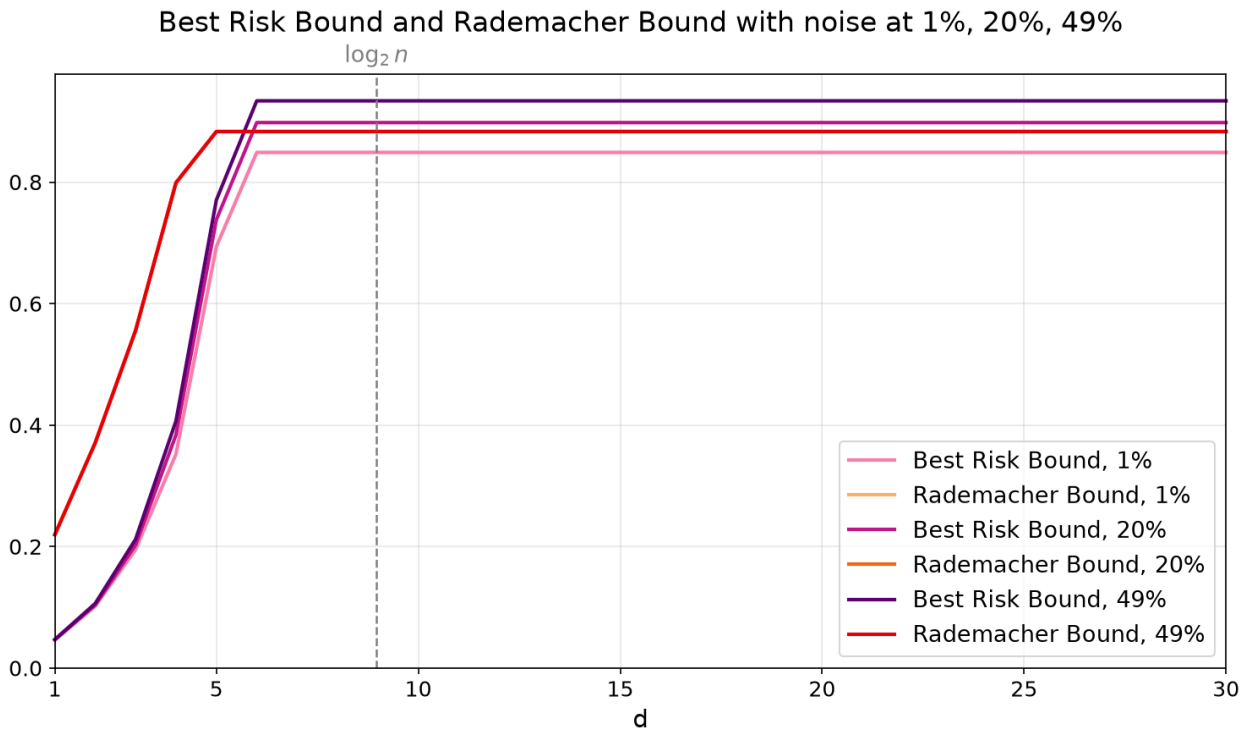


FIGURE 11. Noise and approximation error bound (Theorem 3.1) against Rademacher and Massart bound (Corollary 2.3) for different noise levels (more extreme).

Figure 11 is an attempt to vsiualize better the sensitivity of the noise bound using more extreme noise levels (1%, 20%, 49%), with the same constants as in Figure 10, with poor results.

Appendix A. Fitting constants

Union bound. We just rescale the bound so that it lies in the interval $[0, 1]$. For d large, $\log|\mathscr{H}_d| = N \log 2$. This means that the bound is $2\sqrt{\frac{N \log 2}{2n}}$, hence we need

$$C_{\text{union}} \cdot 2\sqrt{\frac{N \log 2}{2n}} = 1$$

Therefore we set

$$C_{\text{union}} = \frac{1}{2\sqrt{\frac{10^9 \log 2}{2 \cdot 500}}} = \frac{1}{2\sqrt{10^6 \log 2}} = \frac{1}{2\sqrt{\log 2}} 10^{-3} \approx 0.0006$$

Rademacher and Massart bound. Using the code `source/experiments.py`, we print the value of the generalization gap for the highest depth $d = 35$, getting

$$0.2860 \pm 0.0121$$

Therefore, to obtain that the curve of the Rademacher bound saturates at the same level, we need

$$C_{\text{RadMass}} 2\sqrt{2 \log 2} = 0.286 \quad \Rightarrow \quad C_{\text{RadMass}} = \frac{0.286}{2\sqrt{2 \log 2}} = 0.12145302 \approx 0.125$$

which we approximate in eccesso for the bound to lie above the curve.

Noise bound. Similarly, we want the bound to saturate at the same level of the empirical gap curve. The saturating level of the bound, i.e. its value for a larghe depth, is

$$4\sqrt{R^*_{\mathscr{H}_d} M} + 20M \qquad \text{with } M = \frac{(2n+2)\log 2 + 1}{n}$$

hence we set

$$C_{\text{noise}}(4\sqrt{R^*_{\mathscr{H}_d} M} + 20M) = 0.286.$$

We have

$$M = \frac{(1002)\log 2 + 1}{500} \approx 1.3910$$

hence

$$(4\sqrt{R^*_{\mathscr{H}_d} M} + 20M) \approx 4\sqrt{0.2 \cdot 1.391} + 20 \cdot 1.391 \approx 29.9298$$

Thus,

$$C_{\text{noise}} = 0.286/29.9298 \approx 0.01$$

where we use $R^*_d = 0.2$ the Bayes risk, that is the value attained by the best risk for a large enough depth.